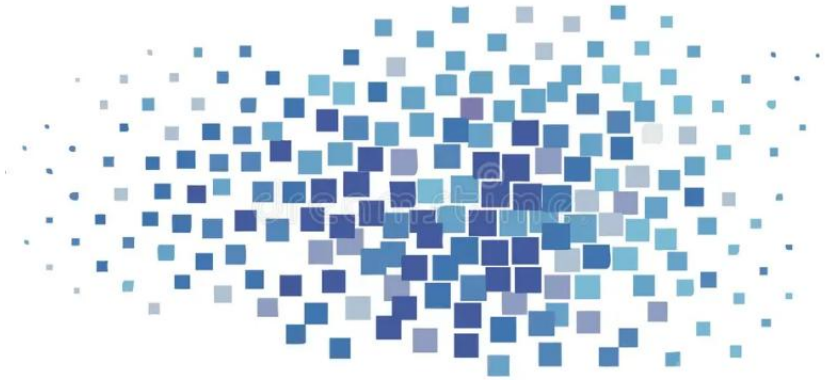
# Information Diffusion in Social Network (cont.)

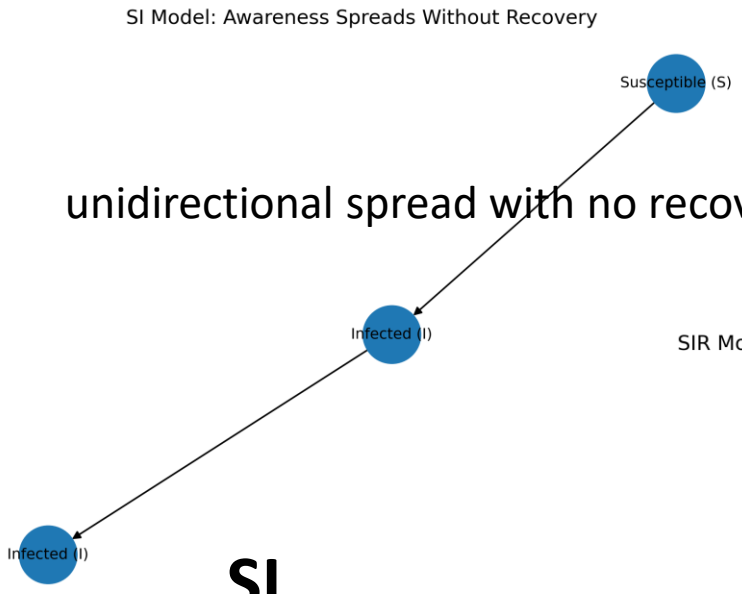
## (SNA – B Tech III CSE)

(Week#6 – Feb 11, 2026)

Dr Dhiren Patel

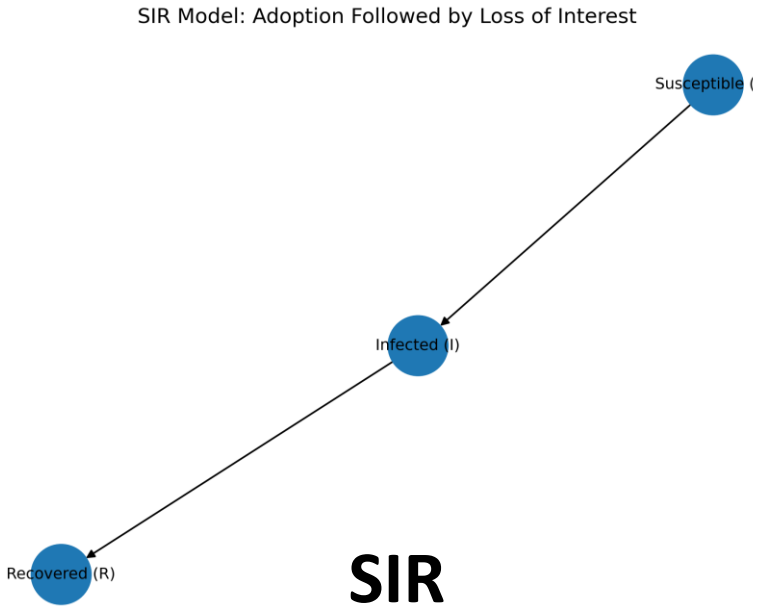


# Visualization

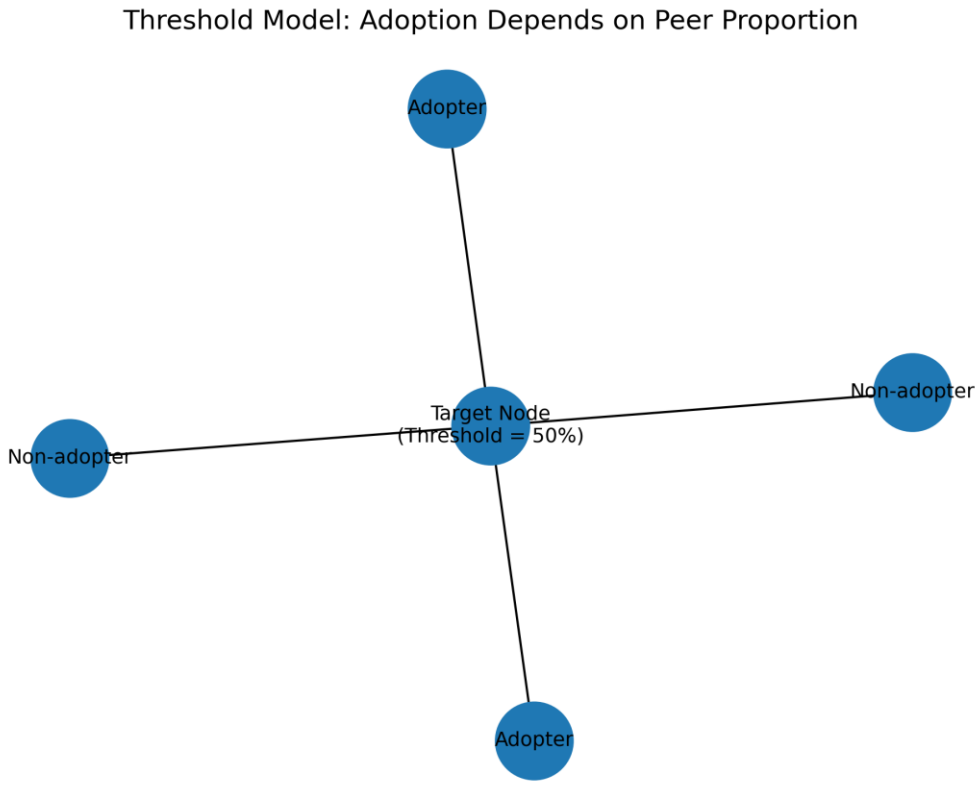



unidirectional spread with no recovery

**SI**

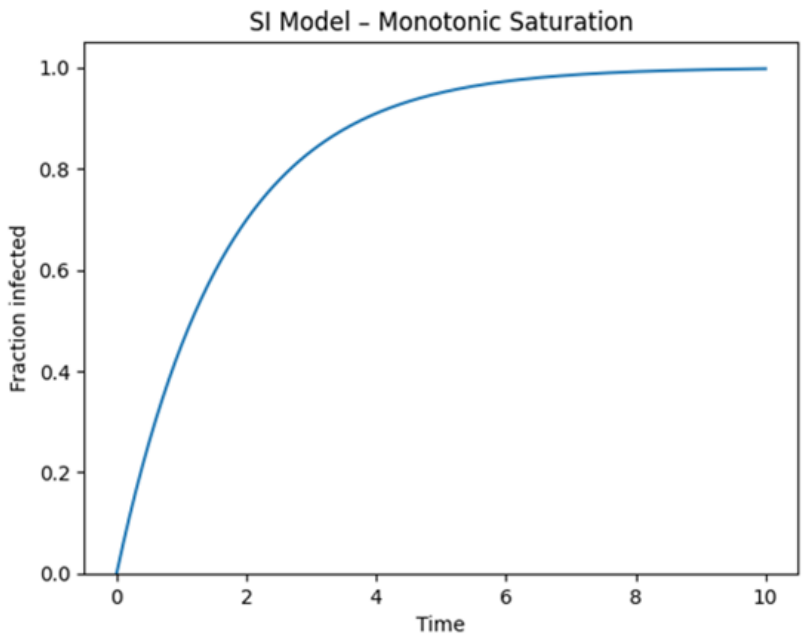


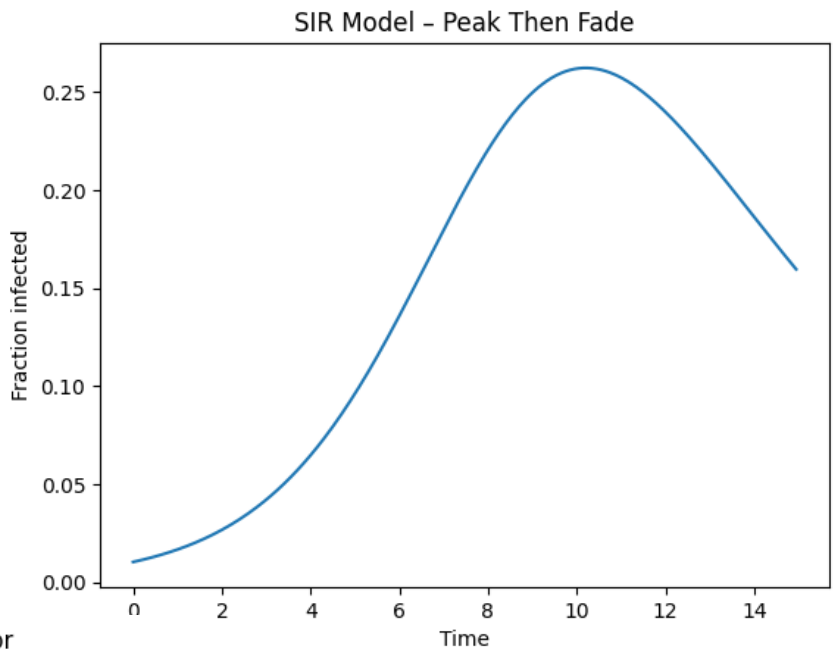

**SIR**

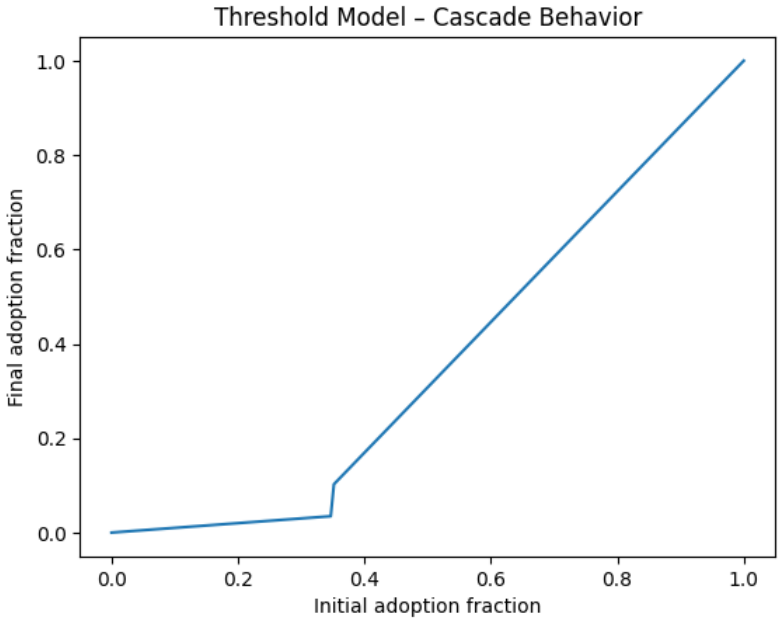



target node adopting only when enough neighbors adopt

**Threshold**

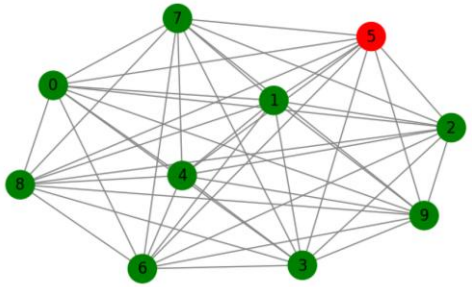

SI Model – Monotonic Saturation
Fraction infected
Time
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10


SIR Model – Peak Then Fade
Fraction infected
Time
0.00
0.05
0.10
0.15
0.20
0.25
0
2
4
6
8
10
12
14

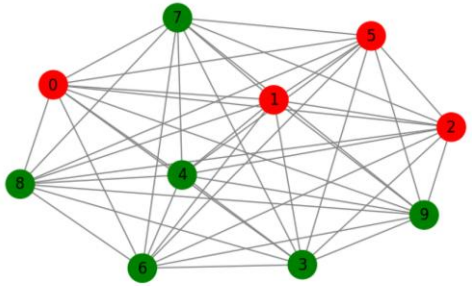

Threshold Model – Cascade Behavior
Final adoption fraction
Initial adoption fraction
0.0
0.2
0.4
0.6
0.8
1.0

# What these models still ignore?

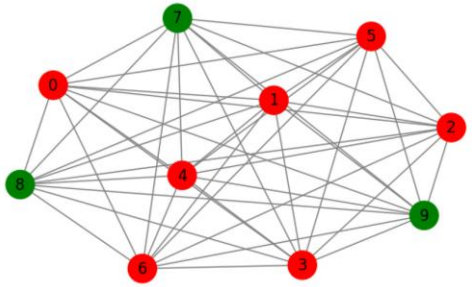
- assumes homogeneous nodes
- ignores network structure
- ignores who is important
- This motivates centrality next.
- And structure must matter….



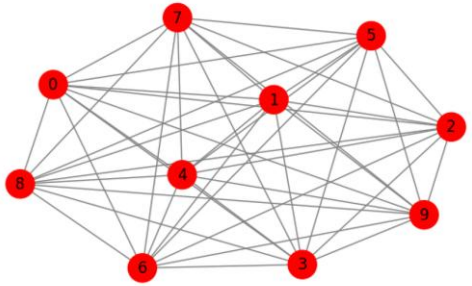
Phase 1



Phase 2



Phase 3



Phase 4

# One Network, Three Behaviors

- **Content**
- Same graph. Different assumptions. Very different results.

| Model | Outcome shape | Real analogy |
|---|---|---|
| SI | monotonic full saturation | rumor |
| SIR | peak then fade | news cycle |
| Threshold | cascade or nothing | protest / tech adoption |

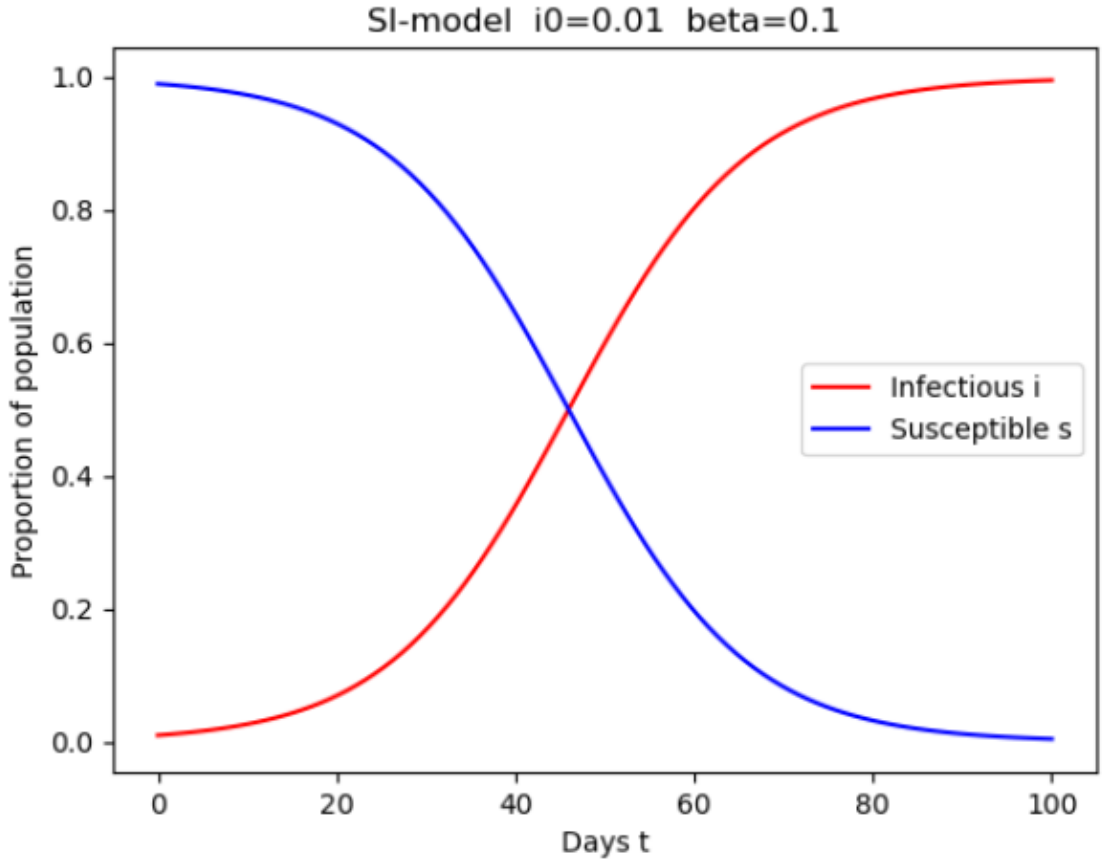
- Structure stays fixed.
- Only behavior rule changes. Dynamics change completely

# Understanding previous slide

- These graphs show: Same nodes, Same edges, Same topology
- Only node colors change over time:
- Green → not adopted, Red → adopted/infected
- So: **structure stays fixed; only behavior rule changes**
- Topology (who connects to whom) vs Dynamics (how influence flows)
- Outcome: same topology + different dynamics ⇒ radically different macro behavior
- If infection never recovers → spread to everyone.
- If recovery exists → peak then die.
- If social proof needed → either nothing or sudden cascade

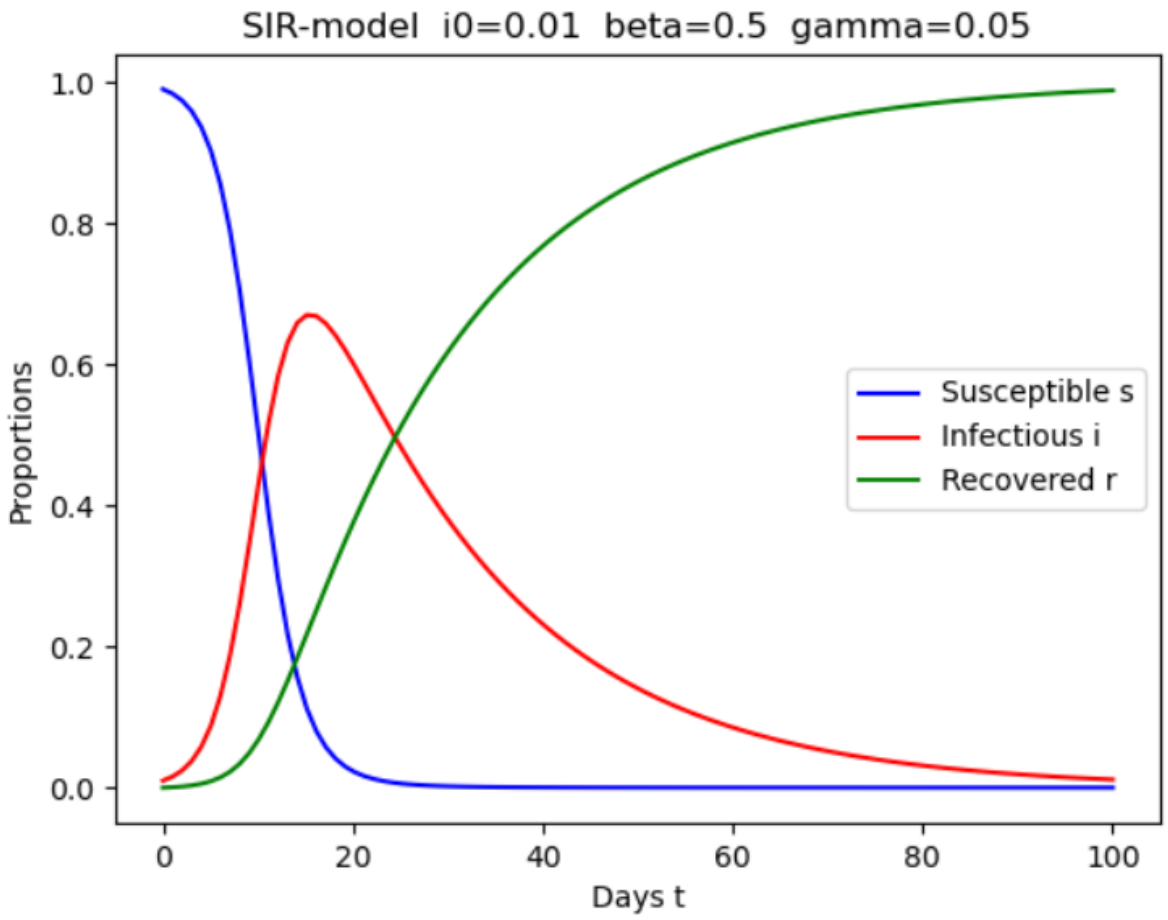
# SI - monotonic full saturation

- SI model - once infected → stays infected
- spreads forever,
- eventually everyone turns red
- Analogy: rumor, malware, gossip
- Nothing ever recovers, so spread never stops



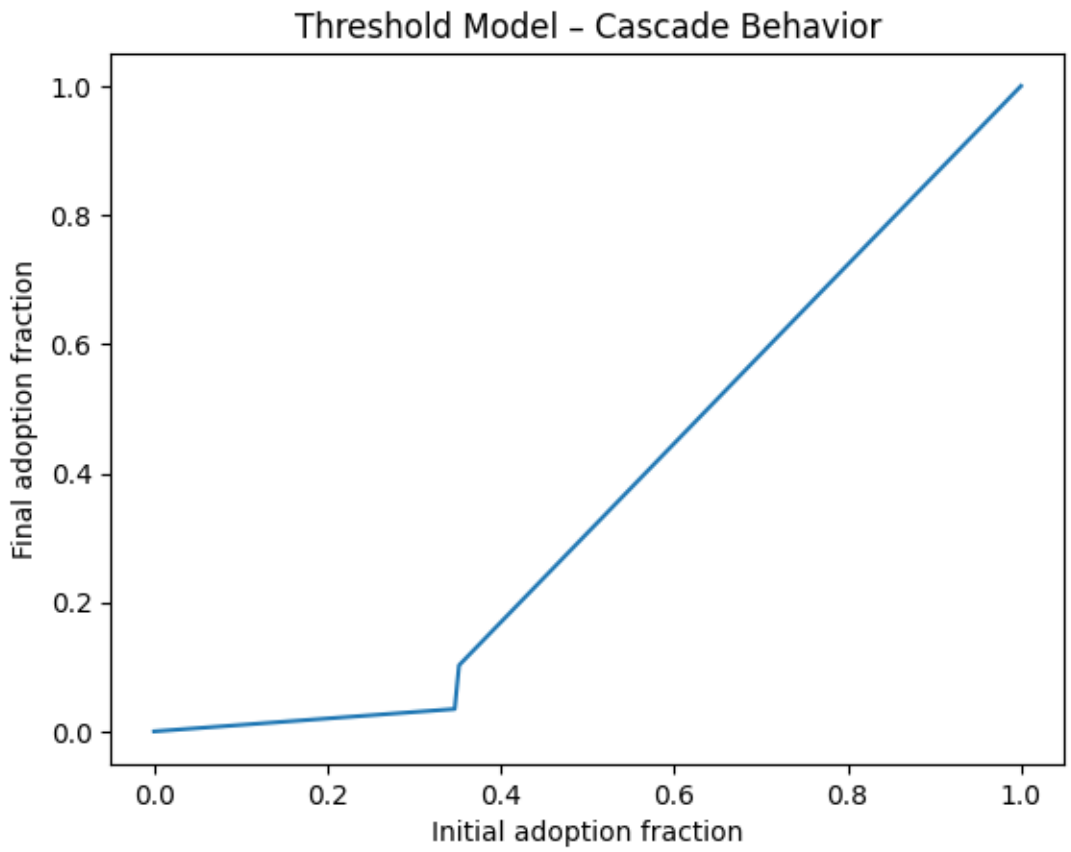
# SIR model - peak then fade

- infected → recover
- recovery removes them from spreading
- spread rises, peaks, then declines
- Analogy: news cycle, disease wave, viral trends
- People lose interest



# Threshold model - cascade or nothing

- adopt only if enough neighbors adopt
- social reinforcement required
- Result: either small local spread, OR sudden large cascade
- No smooth curve.
- Analogy: protests, tech adoption, boycotts
- People wait for social proof.



# Not all diffusion processes behave the same way

| **Real-world behavior** | **Needs which assumption?** |
| --- | --- |
| Rumors, awareness | People never “unlearn” → **SI** |
| Trends, news cycles | People lose interest → **SIR** |
| Peer pressure, protests | People wait for others → **Threshold** |

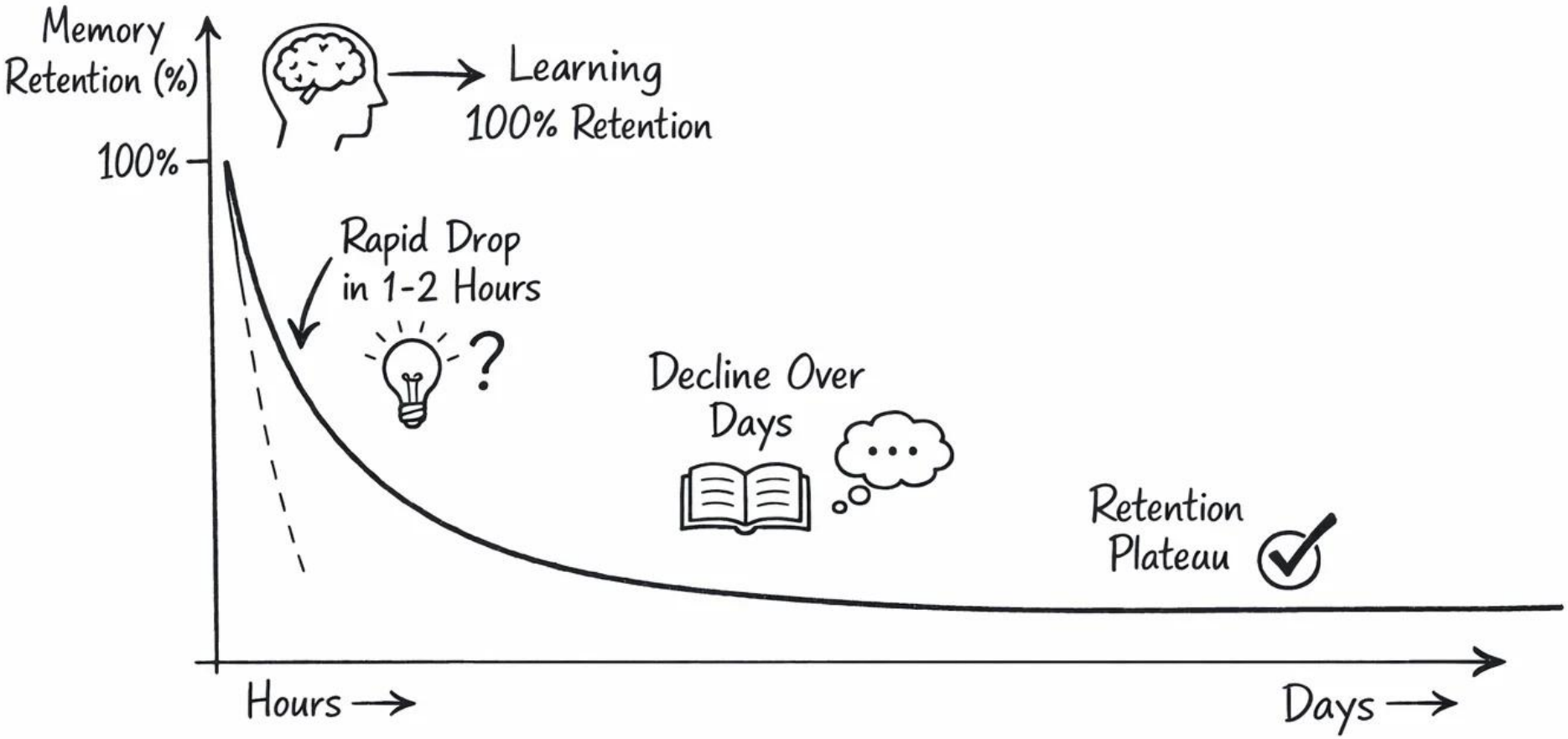
The more social the behavior,
the less SI works and the more Threshold models dominate.

# SIR = Attention Has a Shelf Life

- S decreases, I peaks, R increases
- Bell-shaped infected curve
- spread speed (β) vs forgetting speed (γ)
- **Key equation**
- $R_0 = \frac{\beta}{\gamma}$
- $R_0 > 1 \rightarrow$ viral, $R_0 < 1 \rightarrow$ dies
- **Real examples**
- news cycle, hashtags, exam gossip

# Ebbinghaus Forgetting Curve

Memory Retention Over Time




Structural Learning

# Threshold cascade demo

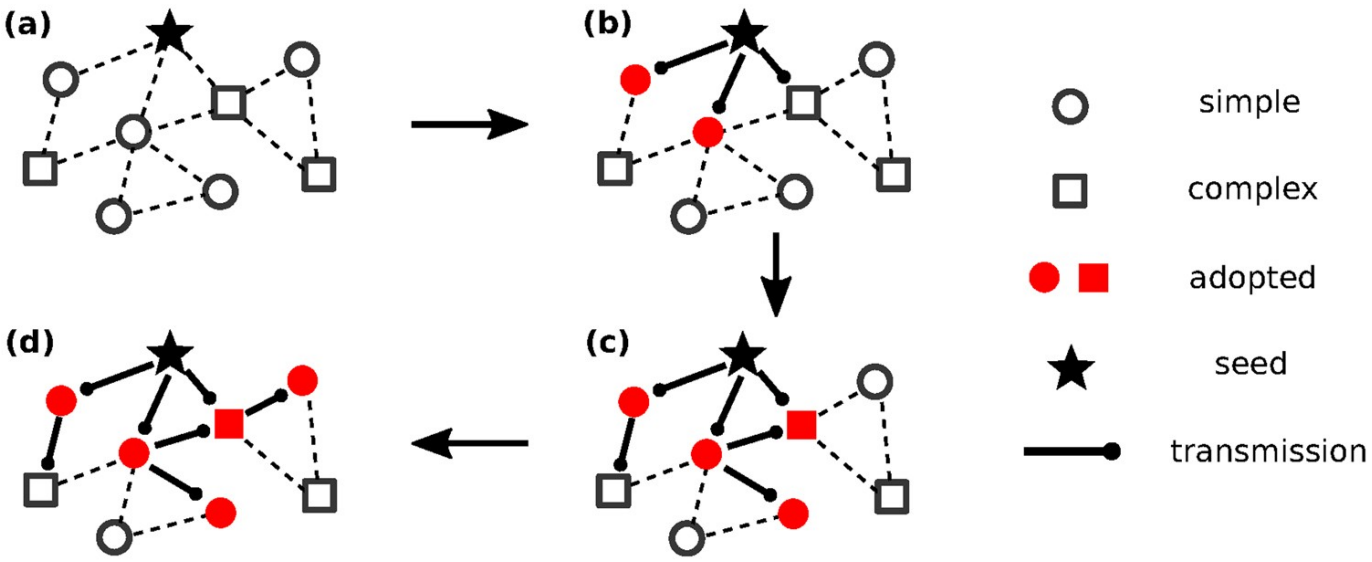
- **Content**
- needs reinforcement
- critical mass exists
- nonlinear (all or nothing)
- **Hook example**
- “Everyone moves to WhatsApp → you also move”

# Behavioral assumptions dominate dynamics more than network structure – e.g. WhatsApp network

- If people never forget → everyone hears it. **(SI Model)**
- If people lose interest → it dies out. **(SIR)**
- If people wait for friends → it may not start at all **(Threshold)**
- **Behavioral assumptions dominate dynamics more than structure.**

(a) Initial state
(b) Early spread
(c) More spread
(d) Saturation

**propagates along edges step-by-step**



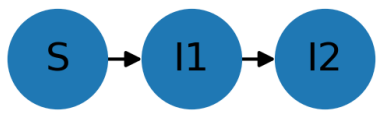
# Cont.

- Humans copy groups, not individuals
- One friend → ignore
  Five friends → maybe
  Everyone → must join
- Local counting (how many neighbors adopted?)
- Personal threshold θ
- All-or-nothing cascade
- Examples – WhatsApp migration, protests, fashion, app adoption, boycotts

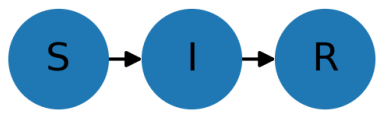
# Cont.

- Threshold Model = Adoption by Social Proof
- people wait for enough neighbors
- each node has θ
- cascades occur after tipping point
- Nonlinear, all-or-nothing
- **Equation**
- $\frac{\text{adopted } neighbors}{degree} \geq \theta$
- **Example**
- “Switching to WhatsApp because all friends moved”

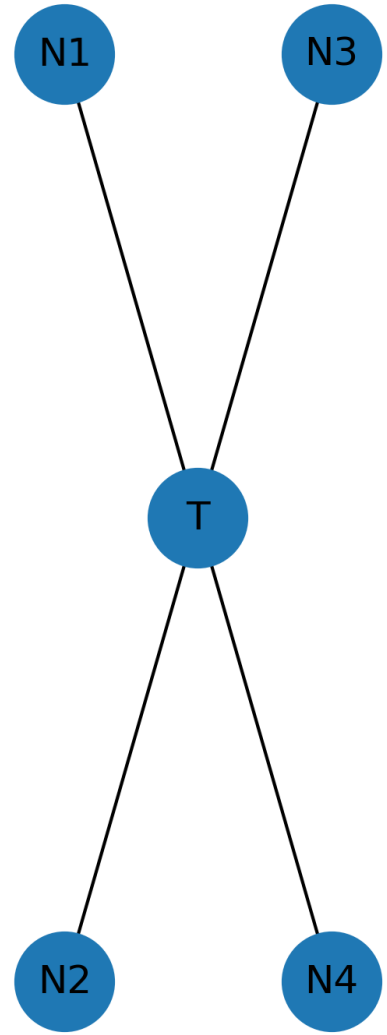
# Evolution of Diffusion Models: Increasing Behavioral Realism




SI Model
(Simple awareness)

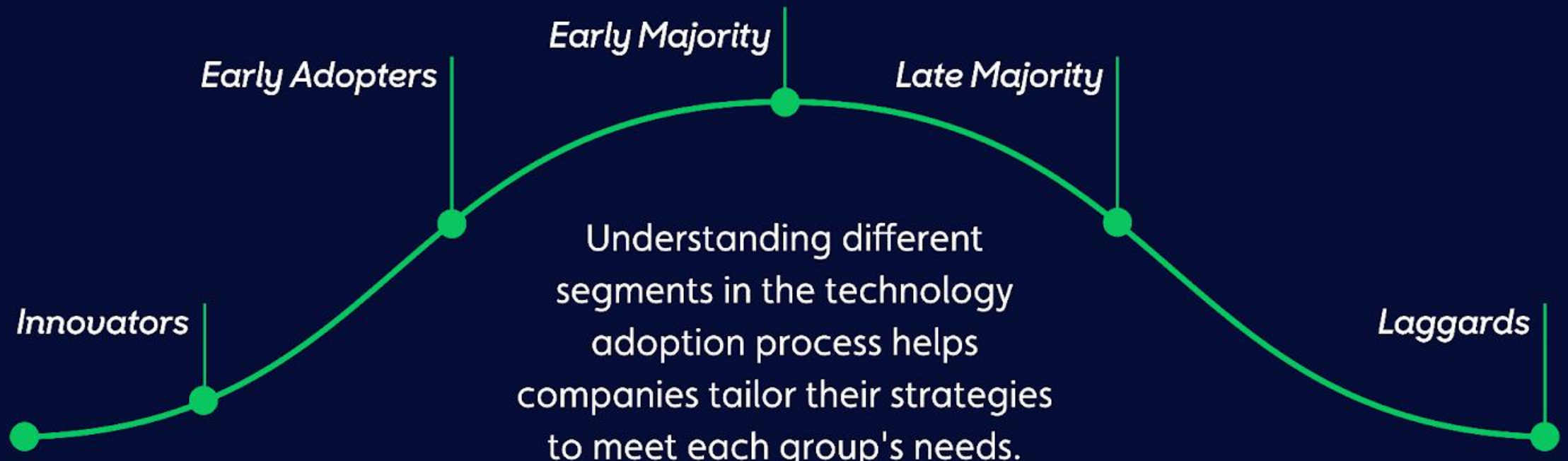



SIR Model
(Attention fades)




Threshold Model
(Social dependence)

# What is the Product Adoption Curve?




Understanding different segments in the technology adoption process helps companies tailor their strategies to meet each group's needs.

# Example – Movie release
## Real-world analogy (intuition only)

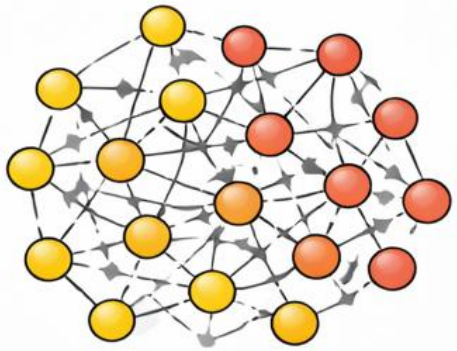
- Teaser (generate interest) – S //S spend, E = Earn (Producer perspective)
- Trailer (generate further interest) - S
- Song(s) (music money) – E
- Promo – The Great Indian Laughter Show, Malls - S
- Premiere – S, Film festival premiere -- S,
- Theatre Release (many screens → business based reduction → few) - E
- OTT Release – E (subscription), OTT - per view premium, OTT World Premiere -- E //Ad revenue
- Re-Release (after big gap, with new climax) -- E

# Concepts and Visualization

- Dense networks → fast diffusion but redundancy
- Sparse networks → slow diffusion
- Presence of hubs → accelerates spread
- Bridges / weak ties → enable cross-community diffusion
- **Structure shapes outcomes more than population size**

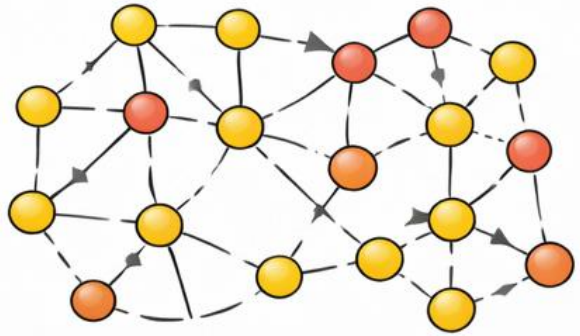
# Network Structure and Diffusion

Same Number of People, Different Connection Patterns → Very Different Outcomes



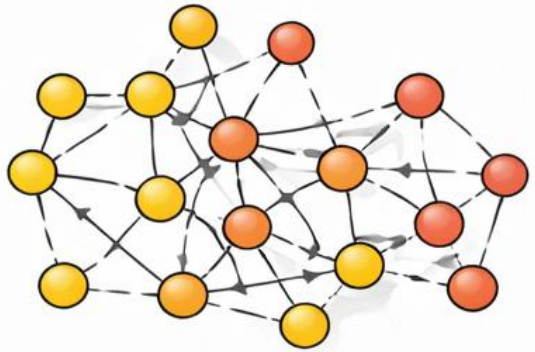
**Dense Network**

*Fast Diffusion, Redundancy*



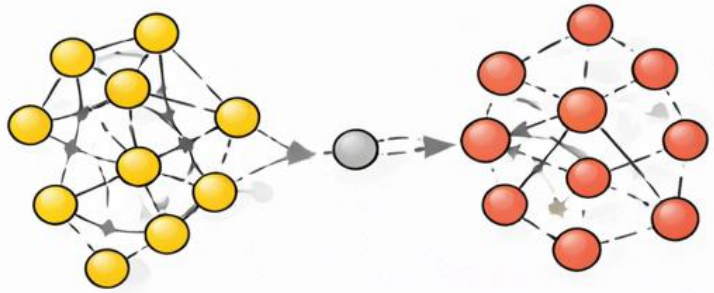
**Sparse Network**

*Slow Diffusion*



**Network with Hubs**

*Accelerated Spread*



**Bridging Network**

*Cross-Community Diffusion*

# Revisiting questions

- Questions –
- “How do we measure who matters?”
- “Which nodes accelerate diffusion?”
- “How do we design interventions?”
- **Some people matter more than others**

# Directives to Answers

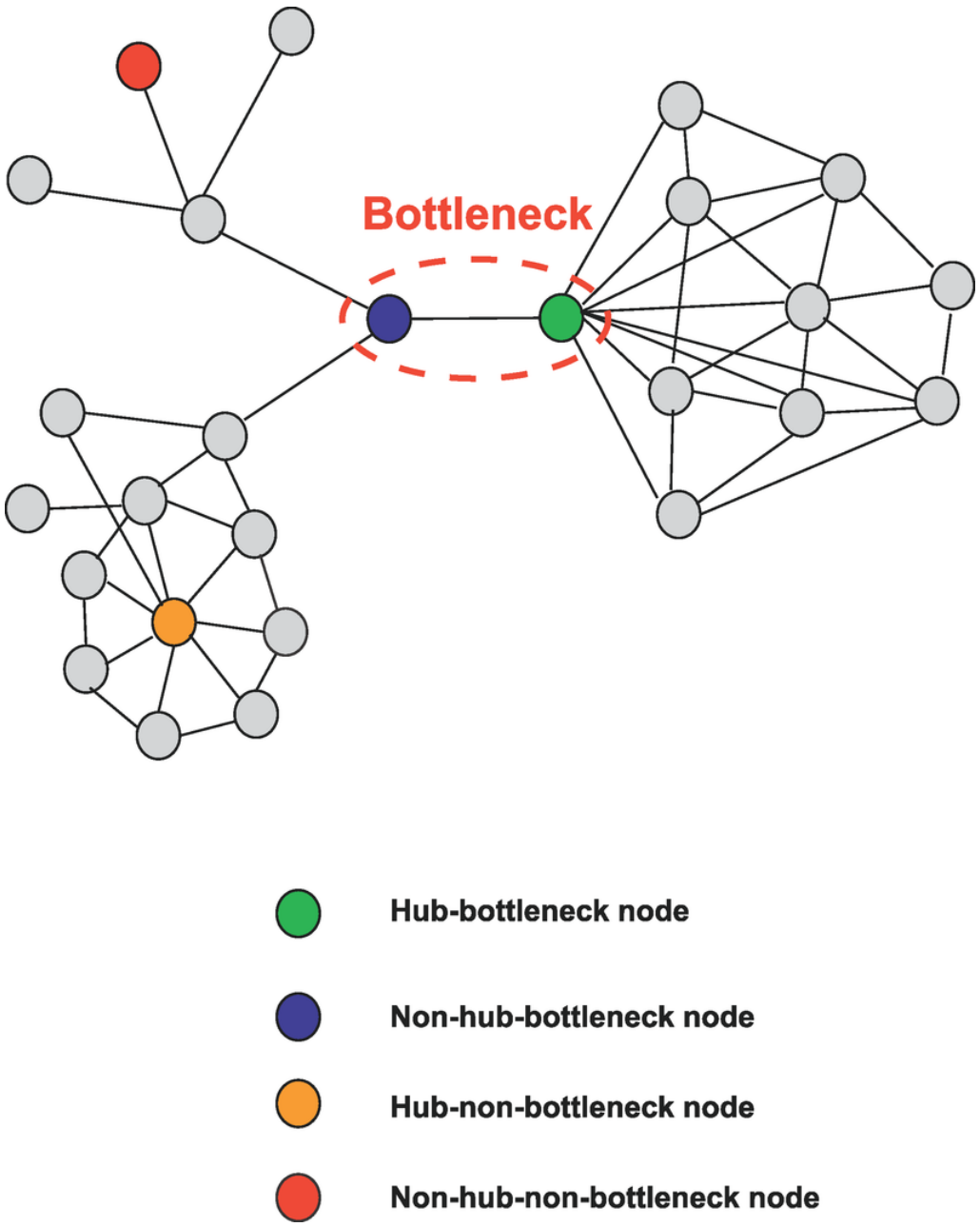
- **Degree centrality -** local reach, good for seeding
- **Betweenness –** bridges, blocking misinformation
- **Closeness -** speed
- **Eigenvector/PageRank -** influence of influencers

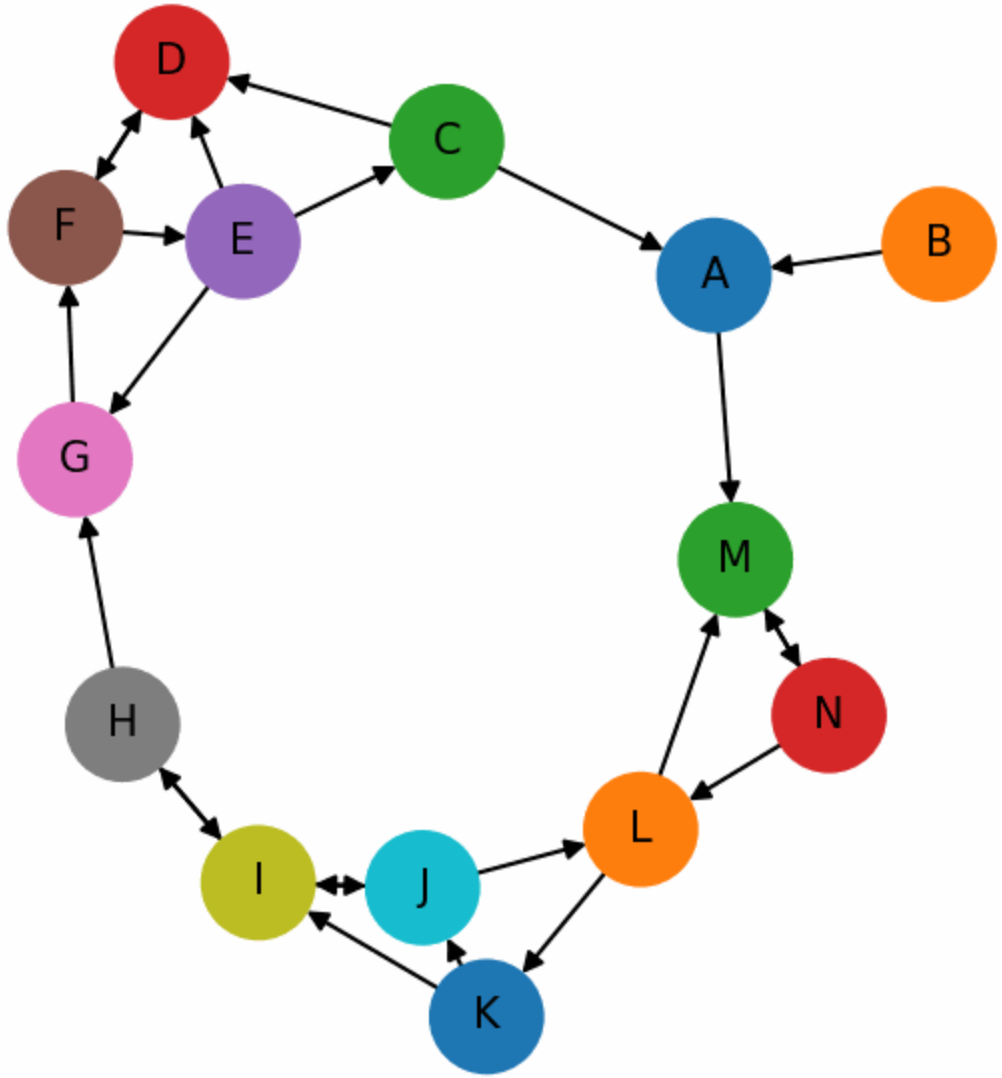
| **Metric** | **Best for** |
|---|---|
| Degree | fast spread |
| Betweenness | containment |
| Closeness | broadcasting |
| Eigenvector | ranking |

# Hub

Star Topology

# Bridge


Bottleneck
Hub-bottleneck node
Non-hub-bottleneck node
Hub-non-bottleneck node
Non-hub-non-bottleneck node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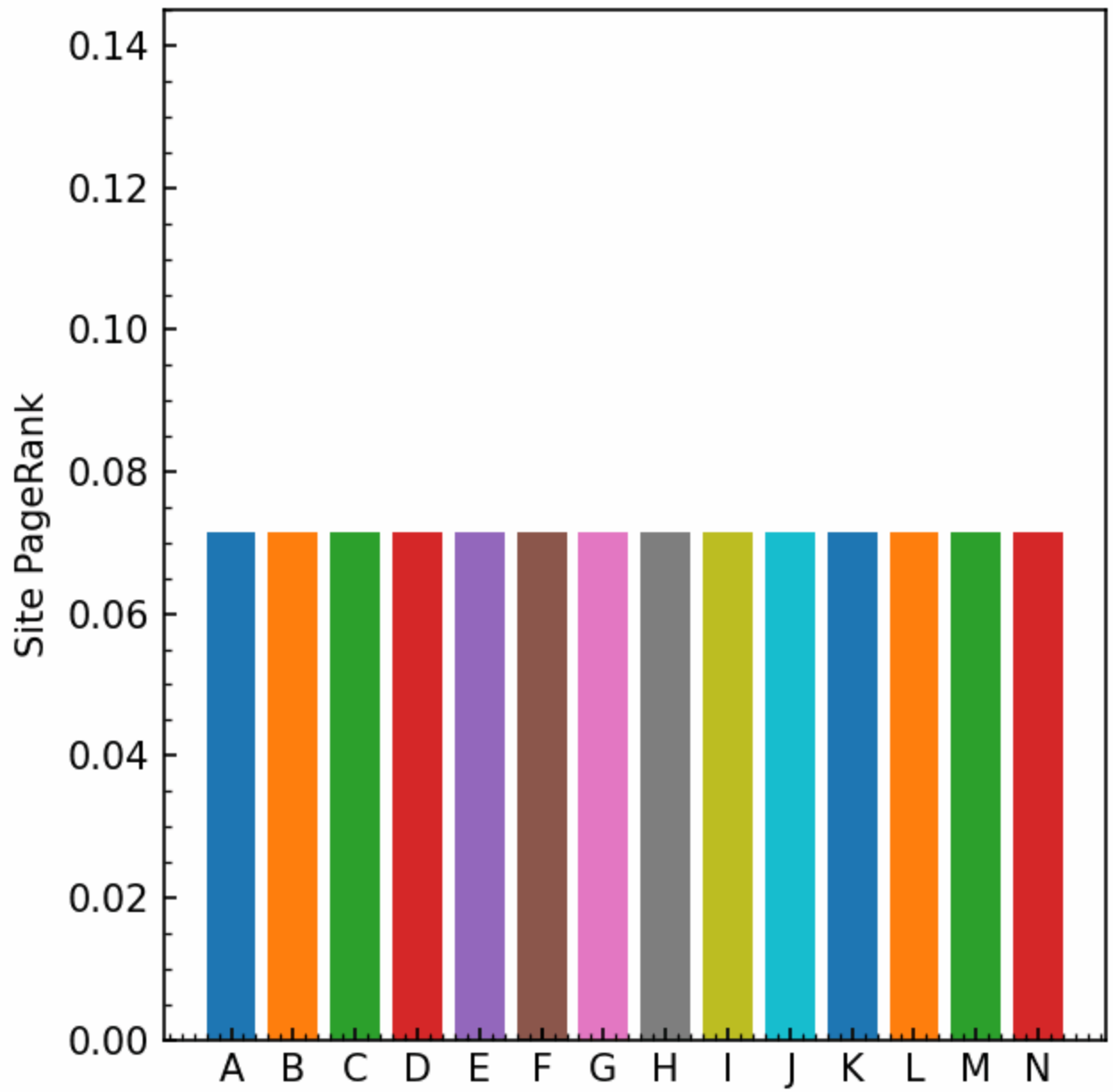
Site PageRank
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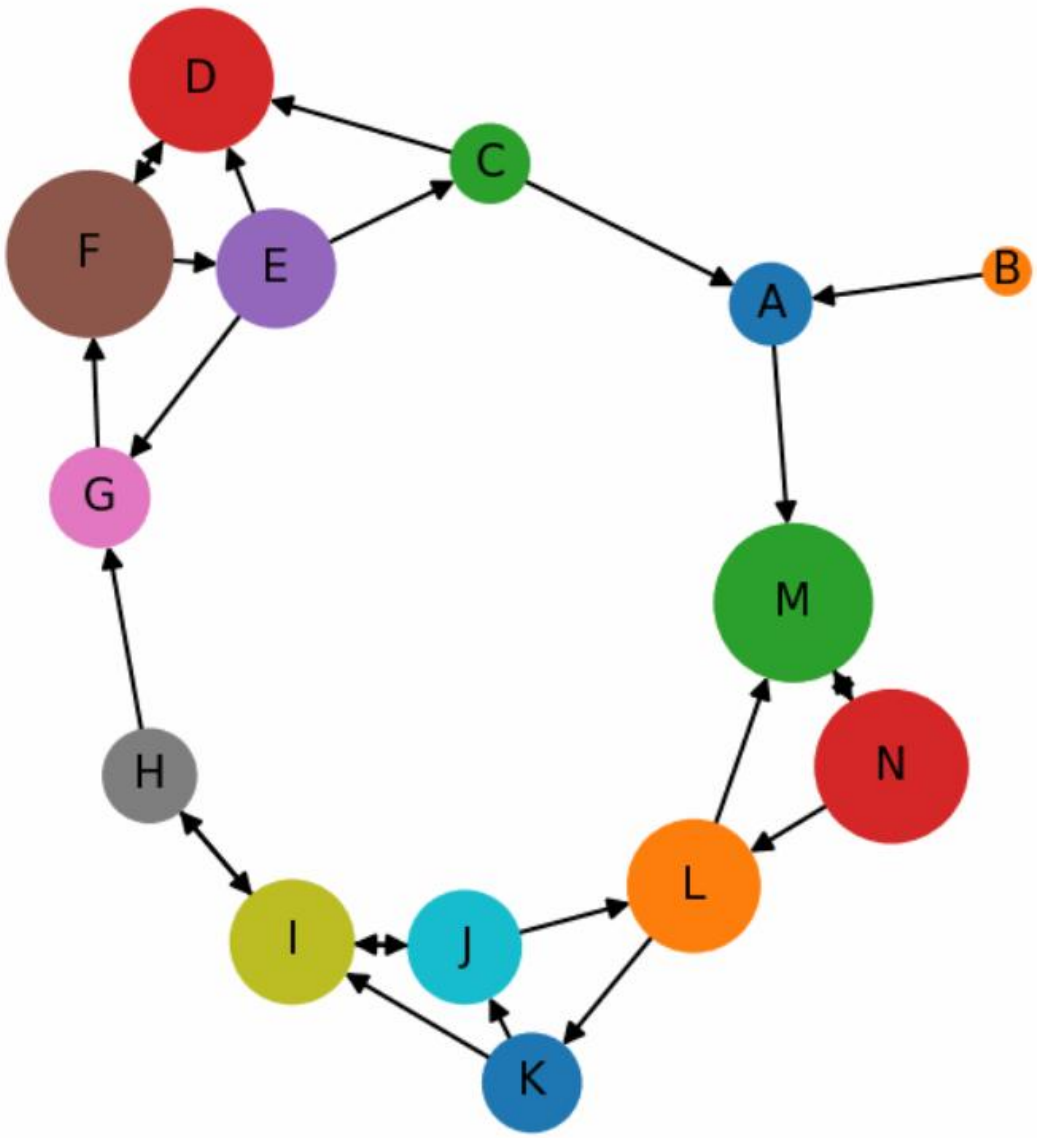
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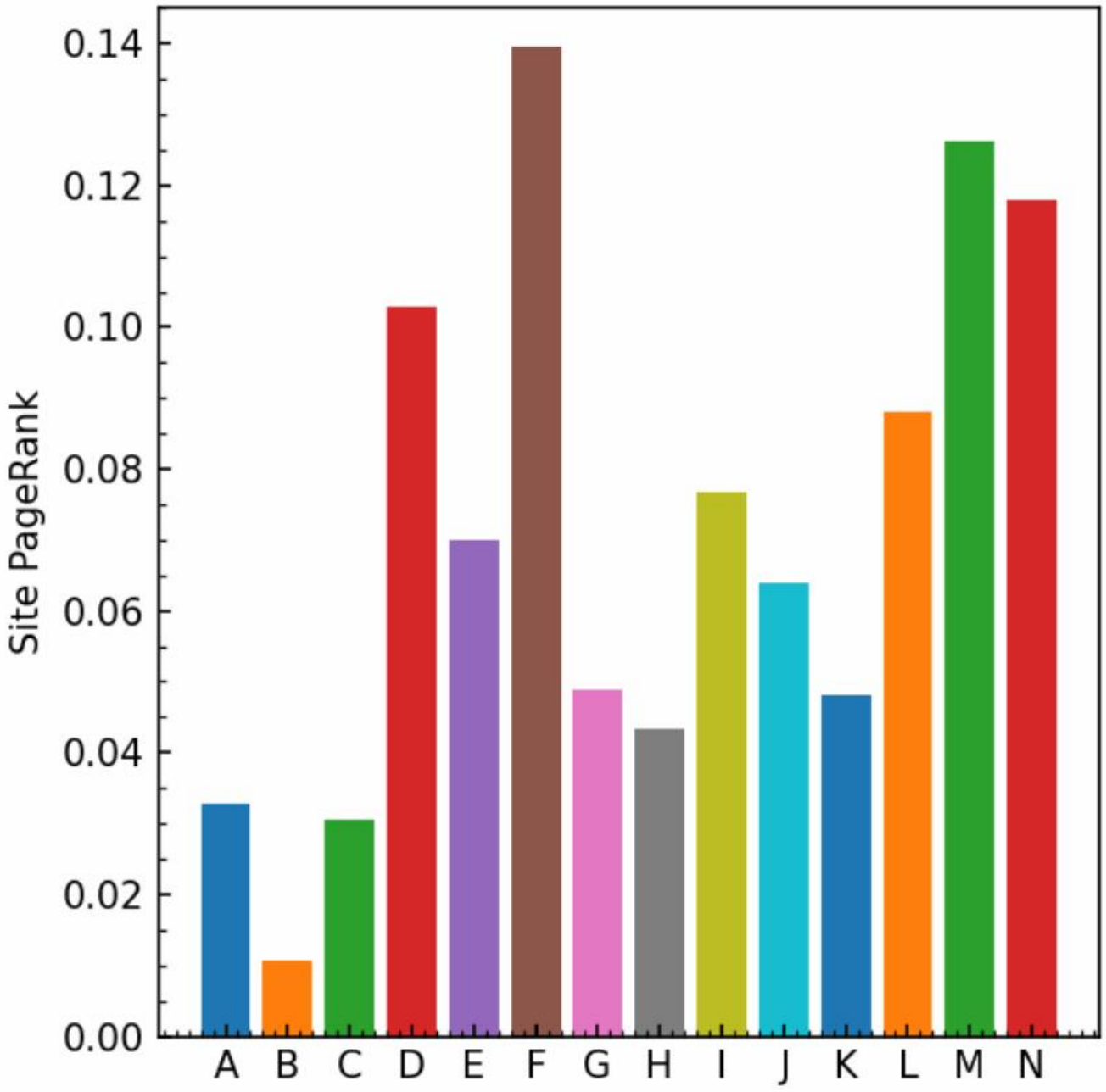
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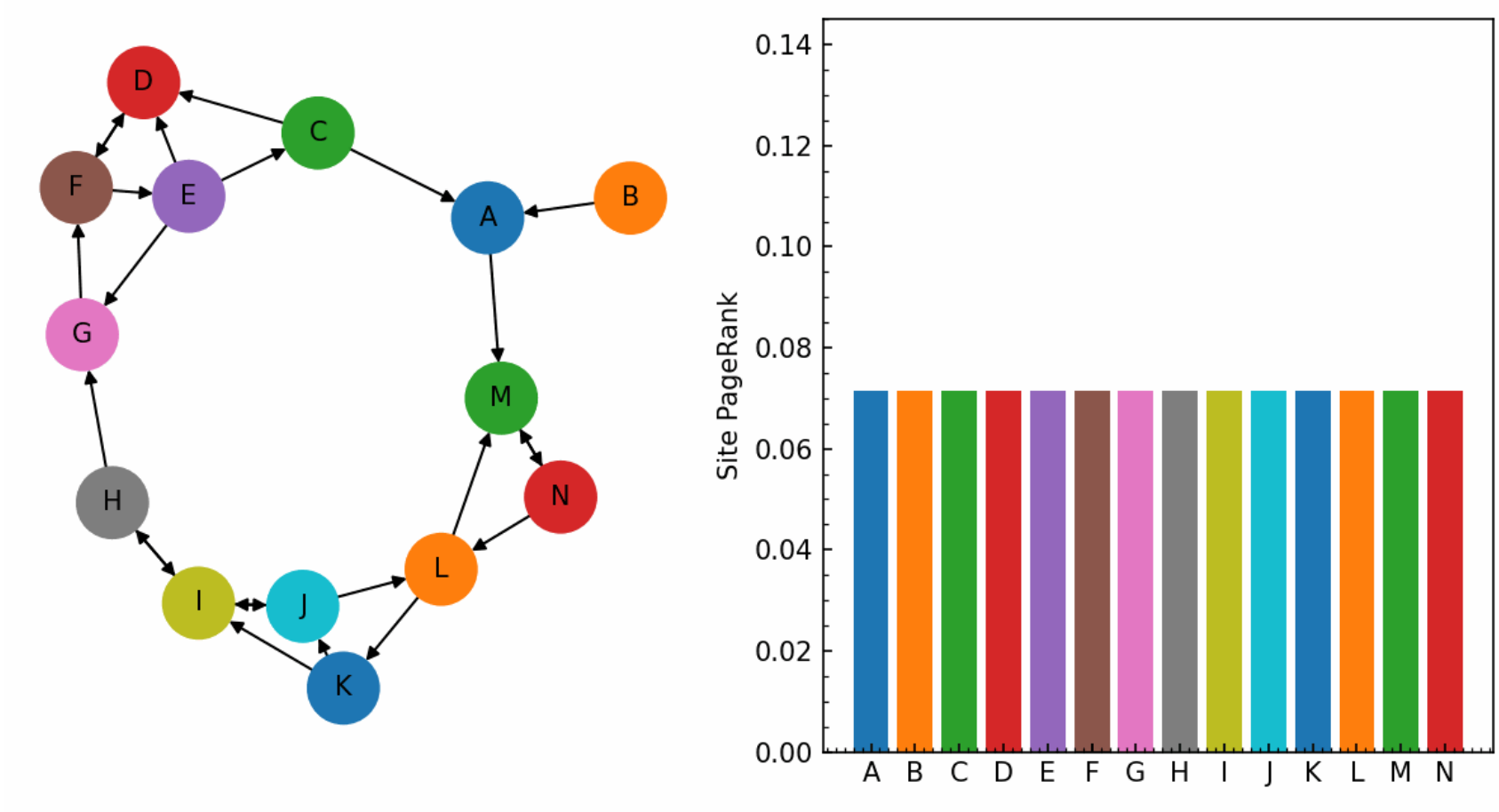
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0.00
Site PageRank
A B C D E F G H I J K L M N

# Explanation

- **Structure (left)**
- directed edges, who points to whom, topology only
- Initially, all nodes have equal rank.
- **Quantitative effect (right)**
- bar heights = PageRank score
- numbers change over iterations, convergence visible
- Now we iteratively redistribute importance along incoming links.
- Notice some nodes grow taller. That’s not because they have many links — it’s because important nodes point to them
- This is exactly eigenvector centrality or PageRank. Importance flows recursively
- **structure → computation → numbers → steady state**
- Scores propagate through links until equilibrium

# Cont.

- Earlier slides –
- **First one - All nodes look the same.**
- Initialization (uniform ranks)
- **Second one - Different ranks**
- After convergence (hierarchy emerges); Some nodes become more important
- We start by assuming everyone equal. Then importance flows through links, and hierarchy emerges automatically

# Eigenvector

- **Eigenvector asks:** “Are my neighbors themselves important?”
- **Degree thinking -** “I know many people.”
- **Eigenvector thinking -** “I know famous/important people.”
- **Example – 1. Hub:** Random person with 100 contacts
- **Example – 2. Eigenvector:** Journalist with 10 contacts:
- Ministers, CEOs, Media heads
- Who has more influence?
- Obviously 2. That’s eigenvector (Eigenvector centrality measures prestige, not popularity)
- hub = count edges → visible
- bridge = sits between groups → visible
- eigenvector = importance of neighbors → invisible